

REDOX EXAMSIDE TRANSFER BOOK

MASTER CANONICAL + CUMULATIVE EDITION

24 uniquely placed eligible PYQs + cumulative mixed challenge + 74-item audit appendix

CANONICAL PRIMARY SETS

RX-ST01 7 • RX-ST02 2 • RX-ST03 4 • RX-ST04 3 • RX-ST05 8 = 24
unique eligible PYQs.

CUMULATIVE SECTION

RX-ST06 reuses selected canonical questions for mixed transfer.
These repetitions do not change primary ownership or coverage counts.

Chapter transfer map

Canonical ownership after full-corpus audit.

MAP

RX-ST01 Oxidation Number

7 PASS

RX-ST02 Oxidation & Reduction

2 PASS

RX-ST03 Oxidising / Reducing Agents

4 PASS

RX-ST04 Redox vs Non-redox

3 PASS

RX-ST05 Types of Redox Reactions

8 PASS

RX-ST06 Cumulative mixed practice

0 new PASS

AUDIT INVARIANT

CHAPTER_ELIGIBLE_TOTAL = 24; CHAPTER_PLACED_UNIQUE = 24; CHAPTER_MISSING = 0; DUPLICATE_PRIMARY = 0.

EXAMSIDE TRANSFER BOOK

SUBTOPIC 01 - OXIDATION NUMBER

Required question set -> concept link -> H1-H3 -> Appendix A -> self-audit

HOW TO USE IT

Attempt first. Use H1 only if stuck; continue to H2/H3 only as needed.

WHY THESE QUESTIONS

These are the canonical primary ExamSIDE transfer items for this subtopic. Other eligible redox questions belong to later subtopics.

Question -> concept map

The required set is frozen before publication.

MAP

ID	Question family	Study concepts	Difficulty
E01	metal carbonyl + ion + neutral	ON-C2/5/6	3
E02	three neutral calculations	ON-C3/5/6	2
E03	variable P states	ON-C5/7	2
E04	calculate then compare jump	ON-C5/6	3
E05	N ordering incl. average/ion	ON-C7/8	2

TRANSFER COVERAGE DENOMINATOR

REQUIRED = 7. These seven questions have canonical primary subtopic RX-ST01. A later-subtopic question is not counted here, preventing both omission and double counting.

PYQ

Try the question first

Find the sum of the metal oxidation states in $\text{Fe}(\text{CO})_5$, VO^{2+} and WO_3 .

FIRST MOVE

Write fixed values and the overall species charge before calculating.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C2 + ON-C5 + ON-C6
Study Guide: rule priority, sum rule, worked routine

CONCEPT HELPER

Use three mini-lanes: fixed special rule $\rightarrow$ ion sum $\rightarrow$ neutral sum.

MISCONCEPTION: Do not read VO^{2+} as VO_2^+ . The 2+ is the charge, not an oxygen subscript.

Progressive help

H1

H1: Solve the three species independently. Do not create one combined equation.

H2

H2: $\text{Fe}(\text{CO})_5$ uses the source's metal-carbonyl rule. VO^{2+} is an ion, so its oxidation-number sum equals +2.

H3

H3: $\text{Fe} = 0$; $\text{V} + (-2) = +2$; $\text{W} + 3(-2) = 0$. Solve V and W, then add.

PYQ

Try the question first

For $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 and K_2FeO_4 , find $x + y + z$ for the three metal oxidation states.

FIRST MOVE

Write fixed values and the overall species charge before calculating.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C3 + ON-C5 + ON-C6
Study Guide: fixed rules, sum rule, worked routine

CONCEPT HELPER

Convert formula $\rightarrow$ number $\rightarrow$ number $\rightarrow$ number. Ordering/adding is the final step.

MISCONCEPTION: Do not use the oxygen subscript as the metal oxidation state.

Progressive help

H1

H1: $\text{K} = +1$ and ordinary $\text{O} = -2$ in all three neutral compounds.

H2

H2: $\text{Cr} = +6$, $\text{Mn} = +7$, $\text{Fe} = +6$. Add only after calculating all three.

PYQ

Try the question first

Find the P oxidation states in $\text{H}_4\text{P}_2\text{O}_7$, $\text{H}_4\text{P}_2\text{O}_5$ and $\text{H}_4\text{P}_2\text{O}_6$.

FIRST MOVE

Write fixed values and the overall species charge before calculating.

OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

ON-C5 + ON-C7
Study Guide: sum rule and variable oxidation states

CONCEPT HELPER

Same element, same method, different oxygen count.

MISCONCEPTION: Do not infer the P value from the formula name or from the O count alone...

Progressive help

H1

H1: All three species are neutral. Use $\text{H} = +1$ and $\text{O} = -2$.

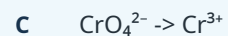
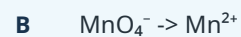
H2

H2: Write three separate sum equations. The values are +5, +3 and +4.

PYQ

Try the question first

Which process changes the tracked element's oxidation state by 5?



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT HELPER

Draw four before $\rightarrow$ after lanes and compare the size of each jump.

MISCONCEPTION: A bigger-looking formula does not mean a bigger oxidation-state change.

Progressive help

H1

H1: Convert the tracked atom in each reactant into an oxidation number before comparing.

H2

H2: Reactant states: Cr +6; Mn +7; Cr +6; C +3.

H3

H3: Product states are +3, +2, +3 and +4. The jump sizes are 3, 5, 3 and 1.

PYQ

Try the question first

Order the nitrogen oxidation states in NO, NO₂, N₂O and NO₃⁻ from highest to lowest.

FIRST MOVE

Write fixed values and the overall species charge before calculating.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C7 + ON-C8
Study Guide: variable states and averages

CONCEPT HELPER

Turn formulas into a number list, then sort the numbers.

MISCONCEPTION: Do not assume N₂O has N = +2 because the formula contains two N atoms.

Progressive help

H1

H1: Calculate each N value before ordering. N₂O gives an average N value.

H2

H2: NO = +2; NO₂ = +4; N₂O = +1 average; NO₃⁻ = +5.

PYQ

Try the question first

Find potassium's oxidation number in K_2O , K_2O_2 and KO_2 .

FIRST MOVE

Write fixed values and the overall species charge before calculating.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C3 + ON-C4
Study Guide: fixed Group 1 rule and oxygen exceptions

CONCEPT HELPER

The oxygen oxidation number changes; potassium does not.

MISCONCEPTION: Do not let peroxide/superoxide exceptions make you change the fixed Gro...

Progressive help

H1

H1: K is Group 1, so $K = +1$ in all three compounds.

PYQ

Try the question first

Arrange N oxidation states in NO, N₂O, NO₂ and N₂O₃ in increasing order.

FIRST MOVE

Write fixed values and the overall species charge before calculating.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C7 + ON-C8
Study Guide: variable states and average values

CONCEPT HELPER

Convert -> compare -> order.

MISCONCEPTION: An average N value is still valid when the question asks for the specie...

Progressive help

H1

H1: Calculate each N value before ordering.

H2

H2: N₂O = +1 average; NO = +2; N₂O₃ = +3 average; NO₂ = +4.

APP A

Use the same reasoning grammar as the Study Guide

READ -> PLACE FIXED VALUES -> SUM -> SOLVE -> CHECK

WHY SOLUTIONS ARE SEPARATE

The question pages are for genuine attempts. Appendix A keeps the full route away from the first view while still giving a verified worked solution for every required item.

APP A

E01 JEE Main 2023 - 8 Apr Evening

Answer: 10

- 1 $\text{Fe}(\text{CO})_5$: Fe = 0 by the source's metal-carbonyl rule.
- 2 VO^{2+} : $V + (-2) = +2$, so $V = +4$.
- 3 WO_3 : $W + 3(-2) = 0$, so $W = +6$.
- 4 Total = $0 + 4 + 6 = 10$.

E02 JEE Main 2020 - 2 Sep Evening

Answer: 19

- 1 $\text{K}_2\text{Cr}_2\text{O}_7$: $2(+1) + 2x + 7(-2) = 0 \rightarrow \text{Cr} = +6$.
- 2 KMnO_4 : $(+1) + x + 4(-2) = 0 \rightarrow \text{Mn} = +7$.
- 3 K_2FeO_4 : $2(+1) + x + 4(-2) = 0 \rightarrow \text{Fe} = +6$.
- 4 Sum = 19.

APP A

E03 JEE Main 2021 - 27 Jul Morning

Answer: +5, +3, +4

- 1 $\text{H}_4\text{P}_2\text{O}_7 \rightarrow \text{P} = +5.$
- 2 $\text{H}_4\text{P}_2\text{O}_5 \rightarrow \text{P} = +3.$
- 3 $\text{H}_4\text{P}_2\text{O}_6 \rightarrow \text{P} = +4.$

E04 JEE Main 2021 - 25 Jul Evening

Answer: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$

- 1 Cr: +6 $\rightarrow$ +3, change 3.
- 2 Mn: +7 $\rightarrow$ +2, change 5.
- 3 Cr: +6 $\rightarrow$ +3, change 3.
- 4 C: +3 $\rightarrow$ +4, change 1.
- 5 Therefore $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}.$

APP A

E05 JEE Main 2021 - 18 Mar Evening

Answer: $\text{NO}_3^- > \text{NO}_2 > \text{NO} > \text{N}_2\text{O}$

- 1 NO: N = +2.
- 2 NO_2 : N = +4.
- 3 N_2O : average N = +1.
- 4 NO_3^- : N = +5.
- 5 Highest to lowest: $\text{NO}_3^- > \text{NO}_2 > \text{NO} > \text{N}_2\text{O}$.

E06 JEE Main 2020 - 7 Jan Morning

Answer: +1, +1, +1

- 1 K is Group 1, so its oxidation number is +1 in K_2O , K_2O_2 and KO_2 .
- 2 The oxygen values differ, but K stays +1.

APP A

E07 JEE Main 2019 - 9 Apr Morning

Answer: $\text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_3 < \text{NO}_2$

- 1 N_2O : average N = +1.
- 2 NO : N = +2.
- 3 N_2O_3 : average N = +3.
- 4 NO_2 : N = +4.
- 5 Increasing order: $\text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_3 < \text{NO}_2$.

Transfer-coverage self-audit

Subtopic -> required questions and question -> support route.

AUDIT

RX-ST01 required set

REQUIRED = 7; PLACED = 7

E01, E02, E03, E04, E05, E06 and E07 are all present exactly once as the canonical primary transfer set.

SUBTOPIC_MISSING

0

WRONG_PLACEMENT

0

CONCEPT_LINK_FAILURES

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

BROKEN_SOURCE_LINKS

0

SCOPE_LEAKS

0

Question -> support route

E01-E04

Each: ELIGIBLE_IN_SCOPE -> RX-ST01 -> stable Study Guide concept link -> correct H1-H3 depth -> misconception/helper -> Appendix A -> source link -> COMPLETE.

E05-E07

Each: ELIGIBLE_IN_SCOPE -> RX-ST01 -> stable Study Guide concept link -> correct H1/H2 depth -> misconception/helper -> Appendix A -> source link -> COMPLETE.

STATUS: PASS

Coverage = $7/7 = 1.00$. No missing required item, duplicate primary placement or scope leak.

EXAMSIDE TRANSFER BOOK

SUBTOPIC 02 - OXIDATION & REDUCTION

Direct-primary electron-loss/gain transfer questions only

REQUIRED SET

Two ExamSIDE questions have this subtopic as their canonical primary reasoning task.

SUPPORT

Both are difficulty 2: H1-H2, concept helper, misconception watch, original source link, and Appendix A solution.

MAP

ID	Question family	Study concept	Difficulty
E01	I ⁻ / S ²⁻ electron counting	OR-C4	2

E02 electron count per CO₂ OR-C4 2

PRIMARY-SUBTOPIC RULE

Other ExamSIDE redox questions may use oxidation-number changes as a prerequisite, but if their primary task is agent identification, redox recognition, reaction classification or balancing, they belong elsewhere.

AUDIT DENOMINATOR

SUBTOPIC_REQUIRED = 2. No later-subtopic question is hidden inside this denominator.

PYQ

Try the question first

X and Y are the electron counts involved when I^- is oxidised to I_2 and S^{2-} is oxidised to S. Find $X + Y$.

FIRST MOVE

Write the oxidation-number change per atom before applying atom count.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

OR-C4
Study Guide: electron jump per atom, then atom count

CONCEPT HELPER

Write two electron-count lanes: $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ and $\text{S}^{2-} \rightarrow \text{S} + 2\text{e}^-$.

MISCONCEPTION

Do not answer $1 + 2$. The iodine product I_2 contains two iodine atoms.

Progressive help

H1

H1: Work per atom first. I changes $-1 \rightarrow 0$; S changes $-2 \rightarrow 0$.

H2

H2: Two I^- form I_2 , so iodine loses 2e^- total. One S^{2-} loses 2e^- .

STOP HERE IF YOU CAN CONTINUE

Appendix A is deliberately separated from the question page.

PYQ

Try the question first

In oxidation of oxalate to CO_2 , how many electrons are involved in producing one CO_2 molecule?

FIRST MOVE

Write the oxidation-number change per atom before applying atom count.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

OR-C4
Study Guide: oxidation-number jump per carbon

CONCEPT HELPER

Do not balance the full permanganate reaction. This question needs only the carbon oxidation-number change.

MISCONCEPTION

Do not report $2e^-$ per CO_2 just because oxalate contains two carbon atoms; it produces two CO_2 molecules.

Progressive help

H1

H1: Find carbon's oxidation state in $\text{C}_2\text{O}_4^{2-}$ and in CO_2 .

H2

H2: Carbon changes $+3 \rightarrow +4$. Each carbon loses $1e^-$; one CO_2 contains one carbon.

STOP HERE IF YOU CAN CONTINUE

Appendix A is deliberately separated from the question page.

APP A

ATTEMPT -> H1 -> H2 -> APPENDIX A

SOLUTION GRAMMAR

Assign the oxidation-number change -> determine e^- per atom -> account for the number of atoms / product molecules.

APP A

E01 JEE Main 2026 - 24 Jan Morning

Answer: 4

- 1 Each I changes $-1 \rightarrow 0$, so each I^- loses $1e^-$.
- 2 Two I^- form I_2 , so $X = 2$ electrons.
- 3 S^{2-} changes $-2 \rightarrow 0$, so $Y = 2$ electrons.
- 4 $X + Y = 4$.

E02 JEE Main 2019 - 10 Jan Evening

Answer: 1 electron per CO_2

- 1 In $C_2O_4^{2-}$, average carbon = $+3$.
- 2 In CO_2 , carbon = $+4$.
- 3 Each carbon is oxidised by one unit, so it loses $1e^-$.
- 4 One CO_2 contains one carbon, hence 1 electron per CO_2 molecule.

Transfer-coverage self-audit

Required questions are accounted for in both directions.

AUDIT

View A - subtopic -> required questions

REQUIRED 2 / PLACED 2

E01 I⁻/S²⁻ electron count

E02 electron count per CO₂ from oxalate

SUBTOPIC_MISSING

0

WRONG_PLACEMENT

0

CONCEPT_LINK_FAILURES

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

BROKEN_SOURCE_LINKS

0

SCOPE_LEAKS

0

CHEMISTRY_ASCII_LEAKS

0

View B - question -> support route

E01

ELIGIBLE_IN_SCOPE -> RX-ST02 -> OR-C4 -> difficulty 2 -> H1-H2 -> helper + misconception -> Appendix A -> source link -> COMPLETE.

E02

ELIGIBLE_IN_SCOPE -> RX-ST02 -> OR-C4 -> difficulty 2 -> H1-H2 -> helper + misconception -> Appendix A -> source link -> COMPLETE.

STATUS: PASS

Coverage = 2/2 = 1.00. No missing primary question, support failure, broken source link, typography leak or scope leak.

EXAMSIDE TRANSFER BOOK

SUBTOPIC 03 - OXIDISING & REDUCING AGENTS

Canonical direct agent questions only - each linked back to the Study Guide

REQUIRED SET

Four direct-primary ExamSIDE questions are included. Questions requiring titration, half-reaction balancing, reagent-specific inorganic knowledge or another topic are not forced into this subtopic.

SUPPORT POLICY

Difficulty 2 -> H1-H2. Difficulty 3 -> H1-H3 plus a concept helper. Every question has a misconception watch and Appendix A solution.

Question -> concept map

Required set fixed before publication.

MAP

ID	Question family	Study concepts	Difficulty
E01	cannot be oxidising agent	AR-C2/C6	2
E02	cannot act both ways	AR-C6	3
E03	H ₂ O ₂ as reducing agent	AR-C2/C6	2

SCOPE FILTER EXAMPLES

Not primary here: 2024 dichromate half-reaction coefficient question (balancing dependency); 2024 Sn/Pb/Tl oxidising-agent count (requires external oxidation-state stability knowledge); 2021 NaH assertion paired with pyridine basicity (mixed outside dependency).

AUDIT DENOMINATOR

SUBTOPIC_REQUIRED = 4.

PYQ

Try the question first

Which species cannot function as an oxidising agent?

A SO_4^{2-}

B MnO_4^-

C N^{3-}

D BrO_3^-

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

AR-C2 + AR-C6

SELF must be reducible; lowest-state check

MISCONCEPTION

Do not choose the species with the largest negative charge by appearance; calculate/recognise the active atom's state.

Progressive help

H1

H1: An oxidising agent must itself undergo reduction - it must be able to gain electrons.

H2

H2: N in N^{3-} is already at -3 in the source-context nitrogen range; it cannot be reduced further.

CONCEPT HELPER

Ask only: can the redox-active atom move to a lower oxidation number?

PYQ

Try the question first

Which compound cannot act both as oxidising and reducing agent?

A H_3PO_4

B H_2SO_3

C H_2O_2

D HNO_2

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

AR-C6
intermediate-state vs highest-state reasoning

MISCONCEPTION

Do not solve this by memorising which reagent is 'strong'; it is an oxidation-state-position question.

Progressive help

H1

H1: A species that can act both ways needs room for its redox-active element to move both up and down in oxidation number.

H2

H2: P in H_3PO_4 is +5. Compare that with the source's maximum-oxidation-number rule for phosphorus.

H3

H3: P(+5) cannot be oxidised further in this source context, so H_3PO_4 cannot act as both an oxidising and a reducing agent.

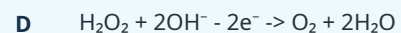
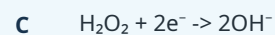
HELPER

Intermediate state $\rightarrow$ can potentially move up or down. Extreme state $\rightarrow$ one direction is blocked.

PYQ

Try the question first

In which given half-reactions does H_2O_2 act as a reducing agent?

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

AR-C2 + AR-C6

RA loses electrons; H_2O_2 oxygen -1 $\rightarrow$ 0

MISCONCEPTION

Do not use the word 'peroxide' to guess the role; H_2O_2 can act either way.

Progressive help

H1

H1: A reducing agent is itself oxidised, so look for H_2O_2 losing electrons.

H2

H2: Reactions 2 and 4 show electron loss and O in H_2O_2 moving -1 $\rightarrow$ 0.

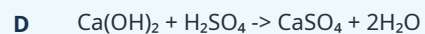
CONCEPT HELPER

Electron on product side / explicit '- e⁻' means the species is losing electrons.

PYQ

Try the question first

Which reaction depicts the oxidising behaviour of H_2SO_4 ?

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

AR-C3 + AR-C7

prove the reagent is reduced in the selected equation

MISCONCEPTION

A reagent known to oxidise in some reactions is not automatically acting as OA in every reaction.

Progressive help

H1

H1: H_2SO_4 is acting as OA only if sulfur in H_2SO_4 is reduced in that equation.

H2

H2: In $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$, S changes +6 $\rightarrow$ +4 while I changes -1 $\rightarrow$ 0.

CONCEPT HELPER

Do not ask 'Is H_2SO_4 an oxidising agent generally?' Ask 'Is it reduced in this particular equation?'

APP A

ATTEMPT -> H1 -> H2 -> H3 (if present) -> APPENDIX A

SOLUTION RULE

Every solution follows the Study Guide pipeline: identify the redox-active atom -> track oxidation number -> decide SELF process -> attach to reactant species -> name OA/RA.

APP A

E01 JEE Main 2024 - 27 Jan Evening

Answer: N^{3-}

- 1 OA must itself be reduced.
- 2 N in N^{3-} is at -3, the lowest source-context nitrogen state.
- 3 It cannot undergo further reduction, so N^{3-} cannot function as an oxidising agent.

E02 JEE Main 2020 - 9 Jan Morning

Answer: H_3PO_4

- 1 P in H_3PO_4 is +5.
- 2 +5 is the maximum source-supported oxidation state for P in this context.
- 3 Therefore H_3PO_4 cannot be oxidised further by phosphorus and cannot show both roles.
- 4 H_2SO_3 , H_2O_2 and HNO_2 contain redox-active atoms at intermediate states.

APP A

E03 JEE Main 2014 - Offline

Answer: 2 and 4

- 1 In H_2O_2 , O = -1.
- 2 When H_2O_2 forms O_2 , O changes -1 $\rightarrow$ 0: oxidation.
- 3 A species that is oxidised acts as the reducing agent.
- 4 Therefore reactions 2 and 4 show H_2O_2 as a reducing agent.

E04 AIEEE 2006

Answer: $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$

- 1 In H_2SO_4 , S = +6.
- 2 In SO_2 , S = +4: sulfur is reduced.
- 3 Therefore H_2SO_4 is the oxidising agent in this reaction.
- 4 HI is oxidised because I changes -1 $\rightarrow$ 0.

Transfer-coverage self-audit

Required questions are counted by canonical primary subtopic.

AUDIT

View A - RX-ST03 -> required questions

REQUIRED 4 / PLACED 4

E01 2024 cannot OA
E02 2020 cannot act both
E03 2014 H₂O₂ as RA
E04 2006 H₂SO₄ oxidising behaviour

SUBTOPIC_MISSING

0

WRONG_PLACEMENT

0

CONCEPT_LINK_FAILURES

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

BROKEN_SOURCE_LINKS

0

SCOPE_LEAKS

0

View B - question -> support route

E01-E02

ELIGIBLE_IN_SCOPE -> RX-ST03 -> Study Guide concept -> difficulty support
-> concept helper/misconception -> Appendix A -> source link -> COMPLETE.

E03-E04

ELIGIBLE_IN_SCOPE -> RX-ST03 -> Study Guide concept -> H1-H2 ->
helper/misconception -> Appendix A -> source link -> COMPLETE.

STATUS: PASS

Coverage = $4/4 = 1.00$. No missing primary question, concept-link failure, hint/solution failure or scope leak.

EXAMSIDE TRANSFER BOOK

SUBTOPIC 04 - REDOX vs NON-REDOX

Direct-primary recognition questions only

REQUIRED SET

Three ExamSIDE questions have redox/non-redox recognition as their canonical primary task.

SUPPORT POLICY

Difficulty 2 -> H1-H2. Difficulty 3 -> H1-H3. Every question includes concept link, helper, misconception watch, source link and Appendix A solution.

Question -> concept map

Freeze the canonical required set before publication.

MAP

ID	Question family	Concept link	Difficulty
E01	recognise redox among neutralisation/allotrope/etc.	RR-C2/C5/C6	2
E02	xenon/O-F oxidation-state recognition	RR-C2/C6 + ON-C4	3

DEFERRED TO NEXT SUBTOPIC

The 2023 $\text{Au}(\text{CN})_2^-/\text{Zn}$ question and 2024/2025 match-the-reaction-type questions are not counted here because their primary task includes redox reaction type classification.

AUDIT DENOMINATOR

SUBTOPIC_REQUIRED = 3. No later classification question is double-counted here.

PYQ

Try the question first

Which option is the redox reaction?

- A H_2SO_4 reacting with NaOH
- B O_2 changing to O_3 in sunlight
- C N_2 combining with O_2 at 2000 K
- D $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ reacting with AgNO_3

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RR-C2 / RR-C5 / RR-C6 - fingerprint, elemental-form trap, MCQ scan.

MISCONCEPTION

Do not call $\text{O}_2 \rightarrow \text{O}_3$ redox: both are elemental oxygen with ON = 0.

Progressive help

H1

H1: Test only oxidation-number change. Neutralisation, allotrope change and precipitation can all be non-redox.

H2

H2: In $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$, N changes $0 \rightarrow +2$ and O changes $0 \rightarrow -2$.

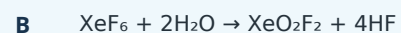
CONCEPT HELPER

Convert each option into a quick ON verdict: unchanged / unchanged / UP+DOWN / unchanged.

PYQ

Try the question first

Which xenon-fluorine reaction is a redox reaction?



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RR-C2 / RR-C6 + ON-C4 - prove change; use O-F oxygen exception.

MISCONCEPTION

Oxygen bonded to fluorine is not -2; the source explicitly gives O = +1 in O_2F_2 .

Progressive help

H1

H1: Track Xe in every option first. In the third option also track oxygen in O_2F_2 .

H2

H2: Xe remains +6 in the first two options and +2 in the fourth. In option 3, Xe changes +4 $\rightarrow$ +6.

H3

H3: In O_2F_2 , O = +1; in O_2 , O = 0. Option 3 therefore has Xe oxidation and O reduction.

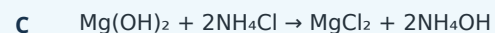
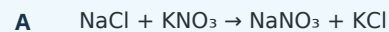
HELPER

Hard-question lane: XeF_4 (+4) $\rightarrow$ XeF_6 (+6) is UP; O_2F_2 oxygen (+1) $\rightarrow$ O_2 (0) is DOWN.

PYQ

Try the question first

Which option is a redox reaction?

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RR-C2 / RR-C6 - scan free-metal change vs ion exchange.

MISCONCEPTION

Double-displacement / ion exchange is not automatically redox.

Progressive help

H1

H1: The first three are ion-exchange patterns. Check whether any oxidation number actually changes.

H2

H2: In option 4, Zn changes $0 \rightarrow +2$ and Ag changes $+1 \rightarrow 0$.

CONCEPT HELPER

Look for a free metal appearing/disappearing: that is a strong cue to test oxidation-number change first.

APP A

ATTEMPT -> H1 -> H2 -> H3 (if present) -> APPENDIX A

SOLUTION GRAMMAR

Track likely atoms -> write oxidation number before -> after -> decide whether any change exists -> verify UP + DOWN for redox.

APP A

E01 JEE Main 2020 - 7 Jan Evening

Answer: $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$

- 1 $\text{H}_2\text{SO}_4 + \text{NaOH}$: no oxidation-number change.
- 2 $\text{O}_2 \rightarrow \text{O}_3$: elemental oxygen remains 0.
- 3 $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$: N 0 $\rightarrow$ +2 and O 0 $\rightarrow$ -2, so redox.
- 4 Cobalt-complex / AgNO_3 reaction is precipitation/ion exchange with no redox change.

E02 JEE Main 2017 - Offline

Answer: $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{XeF}_6 + \text{O}_2$

- 1 A: Xe is +6 in XeF_6 and +6 in XeOF_4 $\rightarrow$ no redox.
- 2 B: Xe is +6 in XeF_6 and +6 in XeO_2F_2 $\rightarrow$ no redox.
- 3 C: Xe +4 $\rightarrow$ +6; O in O_2F_2 +1 $\rightarrow$ 0 $\rightarrow$ redox.
- 4 D: Xe stays +2 and P stays +5 $\rightarrow$ no redox.

APP A

Full reasoning

E03 Zn / Ag redox check

Step 1 - scan A-C.

They are ion-exchange / acid-base style equations. The relevant oxidation numbers stay unchanged.

Step 2 - inspect D.

Zn: $0 \rightarrow +2$ = oxidation.

Ag: $+1 \rightarrow 0$ = reduction.

Step 3 - verdict.

$\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn}(\text{CN})_2$ is redox.

EXAM SCAN LESSON

When several options are simple ion exchanges, a free metal disappearing into a compound while another metal appears in elemental form is a strong cue to test first.

Wrong-option scan



NON-REDOX

No ON change



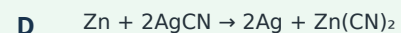
NON-REDOX

No ON change



NON-REDOX

No ON change



REDOX

Zn UP; Ag DOWN

Transfer-coverage self-audit

Required recognition questions accounted for in both directions.

AUDIT

View A - subtopic -> required questions

REQUIRED 3 / PLACED 3

E01 2020 redox recognition
E02 2017 Xe/O-F recognition
E03 2002 redox recognition

SUBTOPIC_MISSING

0

WRONG_PLACEMENT

0

CONCEPT_LINK_FAILURES

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

BROKEN_SOURCE_LINKS

0

SCOPE_LEAKS

0

CHEMISTRY_ASCII_LEAKS

0

View B - question -> support route

E01-E02

ELIGIBLE_IN_SCOPE -> RX-ST04 -> Study Guide concept -> difficulty support
-> helper/misconception -> Appendix A -> source link -> COMPLETE.

E03

ELIGIBLE_IN_SCOPE -> RX-ST04 -> Study Guide concept -> H1-H2 ->
helper/misconception -> Appendix A -> source link -> COMPLETE.

STATUS: PASS

Coverage = $3/3 = 1.00$. No missing primary question, support failure, typography leak or scope leak.

EXAMSIDE TRANSFER BOOK

SUBTOPIC 05 - TYPES OF REDOX REACTIONS

Full-corpus-audited canonical primary set

REQUIRED SET = 8

Five questions from the first build plus three backfilled disproportionation PYQs discovered by the 74-item full-corpus audit.

WHY THE COUNT CHANGED

Duplicate/known reasoning variants do not create new teaching concepts, but they still require deliberate transfer placement. Full-corpus audit corrected the earlier thin denominator.

Question -> concept map

Required set = 8 after full-corpus audit.

MAP

ID	Question family	Concept link	Diff.
E01	2025 type match	RT-C2/C3/C4	2
E02	2024 type match	RT-C2/C3/C4	2
E03	2024 disproportionation discriminator	RT-C4/C5/C6	3
E04	2025 cannot disproportionate	RT-C4 + ON max	2
E05	2023 redox + displacement	RT-C3 + RR-C2	2
E06	2022 NOT disproportionation	RT-C4/C6	2
E07	2022 IS disproportionation	RT-C4/C6	2
E08	2019 example disproportionation	RT-C4/C6	2

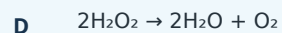
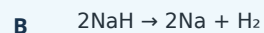
AUDIT DENOMINATOR

SUBTOPIC_REQUIRED = 8. E06-E08 are known/duplicate reasoning variants, but each remains a required canonical PYQ placement.

PYQ

Try the question first

Match four redox reactions with: disproportionation, combination, decomposition and displacement.



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RT-C2 / RT-C3 / RT-C4 - Combination, decomposition, displacement and disproportionation topology.

MISCONCEPTION

H_2O_2 is not only decomposition: O(-1) becomes -2 and 0, so it disproportionates.

Progressive help

H1

H1: Use topology first. NaH starts as one reactant; Ca replaces V; H_2O_2 has O at -1 moving both ways.

H2

H2: A = combination, B = decomposition, C = displacement, D = disproportionation.

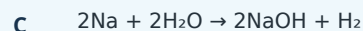
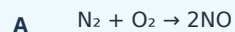
CONCEPT HELPER

Classify by pattern only after checking oxidation-state evidence for the ambiguous H_2O_2 case.

PYQ

Try the question first

Match four reactions with decomposition, displacement, disproportionation and combination.



[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RT-C2 / RT-C3 / RT-C4 - Master classification workflow.

MISCONCEPTION

Product count alone does not prove disproportionation; track the same element's oxidation states.

Progressive help

H1

H1: A combines elements; B is one reactant splitting; C has Na replacing H; D has N(+4) moving to +3 and +5.

H2

H2: A = combination, B = decomposition, C = displacement, D = disproportionation.

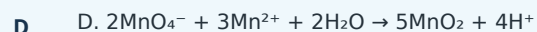
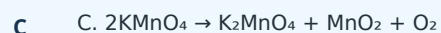
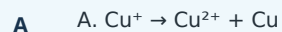
CONCEPT HELPER

The difficult row is D: one N oxidation state (+4) splits lower (+3) and higher (+5).

PYQ

Try the question first

Which listed reactions are disproportionation reactions?



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RT-C4 / RT-C5 / RT-C6 - Split vs converge topology.

MISCONCEPTION

A reaction can contain both oxidation and reduction without disproportionating the same element.

Progressive help

H1

H1: Disproportionation needs one starting ON of the same element to finish both lower and higher.

H2

H2: A: $\text{Cu} +1 \rightarrow +2$ and 0. B: $\text{Mn} +6 \rightarrow +7$ and $+4$. C: $\text{Mn} +7$ only goes down; O is oxidised. D: $\text{Mn} +7$ and $+2$ converge to $+4$.

H3

H3: A and B are split patterns. C is intramolecular/decomposition redox. D is comproportionation.

HELPER

One starting ON splitting = disproportionation; two starting ONs converging = comproportionation.

PYQ

Try the question first

Which chlorine oxyanion does not undergo disproportionation?

A ClO_4^-

B ClO_3^-

C ClO^-

D ClO_2^-

OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RT-C4 + ON maximum-state check - Disproportionation needs room to move both lower and higher.

MISCONCEPTION

Having several oxidation states does not mean every one of them can disproportionate.

Progressive help

H1

H1: A highest oxidation state cannot be oxidised further, so it cannot split both upward and downward.

H2

H2: Cl in ClO_4^- is +7, the highest source-supported state for chlorine.

CONCEPT HELPER

Disproportionation needs an intermediate state with a possible higher and lower state.

PYQ

Try the question first

For $2[\text{Au}(\text{CN})_2]^- + \text{Zn} \rightarrow 2\text{Au} + [\text{Zn}(\text{CN})_4]^{2-}$, which labels are correct: redox, displacement, decomposition, combination?

CLASSIFICATION TARGET

Decide redox status and reaction topology separately; more than one label can be correct.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RT-C3 + RR-C2 - A reaction can carry both redox and displacement labels.

MISCONCEPTION

Redox and displacement are not mutually exclusive labels.

Progressive help

H1

H1: Track Zn and Au: Zn $0 \rightarrow +2$; Au $+1 \rightarrow 0$.

H2

H2: The reaction is redox, and Zn displaces Au. It is not decomposition or combination.

CONCEPT HELPER

Separate the two questions: Is there ON change? What is the reaction topology?

PYQ

Try the question first

Which of the given reactions is NOT an example of disproportionation reaction?

- A $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
- B $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_3 + \text{HNO}_2$
- C $\text{MnO}_4^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$
- D $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RT-C4 / RT-C6 - split-vs-converge / same-element topology.

CONCEPT HELPER

One starting ON must split into two product ONs on opposite sides.

Progressive help

H1

Disproportionation requires the SAME element to start at one oxidation state and finish both higher and lower.

H2

In option C, Mn only goes $+7 \rightarrow +4$; the electron is supplied externally. There is no same-element split.

STOP CHECK

If the split pattern is now clear, finish independently before Appendix A.

PYQ

Try the question first

Which one of the following is an example of disproportionation reaction?

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RT-C4 / RT-C6 - split-vs-converge / same-element topology.

CONCEPT HELPER

+6 $\rightarrow$ +7 and +4 is the split pattern.

Progressive help

H1

Look for one starting oxidation state of the same element producing both a higher and a lower state.

H2

In A, Mn starts at +6 in MnO_4^{2-} and forms Mn(+7) and Mn(+4).

STOP CHECK

If the split pattern is now clear, finish independently before Appendix A.

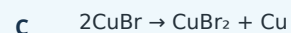
PYQ

Try the question first

Which option is an example of a disproportionation reaction?



B permanganate decomposition option



D permanganate + iodide redox option

OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RT-C4 / RT-C6 - split-vs-converge / same-element topology.

CONCEPT HELPER

Same Cu(+1) split: one part rises, one part falls.

Progressive help

H1

Track Cu in CuBr. Cu starts at +1.

H2

In $2\text{CuBr} \rightarrow \text{CuBr}_2 + \text{Cu}$, Cu(+1) becomes Cu(+2) and Cu(0).

STOP CHECK

If the split pattern is now clear, finish independently before Appendix A.

APP A

ATTEMPT → H1 → H2 → H3 (if present) → APPENDIX A

SOLUTION GRAMMAR

Prove redox if needed -> inspect topology -> track same-element oxidation states for disproportionation/comproportionation -> state every valid reaction label.

APP A

E01 JEE Main 2025 - 28 Jan Morning

Answer: A-II, B-III, C-IV, D-I

- 1 A is combination.
- 2 B is decomposition.
- 3 C is displacement.
- 4 D is disproportionation because O(-1) becomes -2 and 0.
- 5 Match: A-II, B-III, C-IV, D-I.

E02 JEE Main 2024 - 6 Apr Evening

Answer: A-IV, B-I, C-II, D-III

- 1 A is combination.
- 2 B is decomposition.
- 3 C is displacement.
- 4 D is disproportionation: N(+4) becomes +3 and +5.
- 5 Match: A-IV, B-I, C-II, D-III.

APP A

E03 JEE Main 2024 - 1 Feb Morning

Answer: A and B

- 1 A: $\text{Cu}(+1)$ becomes $+2$ and $0 \rightarrow$ disproportionation.
- 2 B: $\text{Mn}(+6)$ becomes $+7$ and $+4 \rightarrow$ disproportionation.
- 3 C: Mn only decreases while O is oxidised.
- 4 D: $\text{Mn}(+7)$ and $\text{Mn}(+2)$ converge to $+4 \rightarrow$ comproportionation.
- 5 Therefore A and B.

E04 JEE Main 2025 - 22 Jan Evening

Answer: ClO_4^-

- 1 Cl in ClO_4^- is $+7$.
- 2 Disproportionation requires both oxidation and reduction of the same starting state.
- 3 At $+7$ chlorine cannot be oxidised further in the source framework.
- 4 Therefore ClO_4^- does not disproportionate.

APP A

E05 JEE Main 2023 - 6 Apr Morning

Answer: Redox and displacement only

- 1 Zn: $0 \rightarrow +2$, so oxidation.
- 2 Au: $+1 \rightarrow 0$, so reduction. Therefore redox.
- 3 Zn replaces Au from the complex species, so displacement.
- 4 Not decomposition or combination.
- 5 Correct: redox and displacement only.

APP A

E06 JEE Main 2022 - 26 Jul Morning

Answer: Option C

- 1 A: O in H_2O_2 is -1 and becomes -2 and 0 $\rightarrow$ disproportionation.
- 2 B: N in NO_2 is +4 and becomes +5 and +3 $\rightarrow$ disproportionation.
- 3 C: Mn +7 $\rightarrow$ +4 only; this is a reduction half-reaction, not disproportionation.
- 4 D: Mn +6 $\rightarrow$ +7 and +4 $\rightarrow$ disproportionation.

E07 JEE Main 2022 - 26 Jun Evening

Answer: Option A

- 1 A: Mn +6 $\rightarrow$ +7 and +4, so the same starting state is oxidised and reduced.
- 2 B: Mn is only reduced.
- 3 C and D are ordinary intermolecular redox reactions with different species carrying oxidation/reduction.
- 4 Therefore A is disproportionation.

APP A

E08 JEE Main 2019 - 12 Apr Morning

Step 1: Cu in CuBr is +1.

Step 2: In CuBr₂, Cu = +2; elemental Cu = 0.

Step 3: The same Cu(+1) starting state becomes one higher state and one lower state.

Verdict: $2\text{CuBr} \rightarrow \text{CuBr}_2 + \text{Cu}$ is a disproportionation reaction.

AUDIT

View A - RX-ST05 -> required questions

REQUIRED 8 / PLACED 8

E01-E05 original canonical set.
E06-E08 backfilled from the 74-item corpus audit.
Missing = 0.

SUBTOPIC_MISSING

0

WRONG_PLACEMENT

0

CONCEPT_LINK_FAILURES

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

BROKEN_SOURCE_LINKS

0

SCOPE_LEAKS

0

DUPLICATE_PRIMARY

0

View B - backfill trace

E06

M24 -> ELIGIBLE_IN_SCOPE -> RX-ST05 -> RT-C4/C6 -> H1-H2 -> Appendix A -> source link -> COMPLETE.

E07-E08

M25/M34 -> ELIGIBLE_IN_SCOPE -> RX-ST05 -> RT-C4/C6 -> H1-H2 -> Appendix A -> source link -> COMPLETE.

STATUS: PASS

Coverage = $8/8 = 1.00$. The full-corpus audit now closes the subtopic denominator.

EXAMSIDE CUMULATIVE CHALLENGE

SUBTOPIC 06 - INTEGRATED DECISION PIPELINE

Six previously canonical questions reused only for cumulative transfer - no duplicate primary placement

WHY REUSE?

The purpose is integration: the learner must choose the correct stopping point of P1-P7 across mixed question families.

AUDIT RULE

Every question keeps its original primary subtopic. This cumulative book creates no new primary coverage count and cannot hide or duplicate canonical placement.

Cumulative question -> primary home

Mixed practice without corrupting the coverage ledger.

MAP

ID	Primary home	Concept link	Difficulty
C01	RX-ST01	ON-C3 / ON-C5 / ON-C6	2
C02	RX-ST02	OR-C4	2
C03	RX-ST03	AR-C2 / AR-C6	2
C04	RX-ST04	RR-C2 / RR-C6 + ON-C4	3
C05	RX-ST05	RT-C4 / RT-C5 / RT-C6	3

LEDGER RULE

NEW_PRIMARY_REQUIRED = 0. These six questions were already placed in RX-ST01-05. Here they are a cumulative sample only; primary ownership remains unchanged.

PYQ

Try first - choose your pipeline stopping point

For $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 and K_2FeO_4 , find the sum of the three metal oxidation states.

FIRST MOVE

Decide whether this needs only oxidation-number work, electron accounting, agent logic, or full type classification.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

ON-C3 / ON-C5 / ON-C6 | Primary home: RX-ST01

PIPELINE HELPER

This tests P2 ASSIGN only; do not run the full redox pipeline.

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help

H1

H1: $\text{K} = +1$ and ordinary $\text{O} = -2$ in all three neutral compounds.

H2

H2: $\text{Cr} = +6$, $\text{Mn} = +7$, $\text{Fe} = +6$. Add only after calculating all three.

STOP CHECK

If H2 makes the representation clear, finish independently before looking at Appendix A.

PYQ

Try first - choose your pipeline stopping point

X and Y are electron counts when I^- is oxidised to I_2 and S^{2-} is oxidised to S. Find $X + Y$.

FIRST MOVE

Decide whether this needs only oxidation-number work, electron accounting, agent logic, or full type classification.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)**CONCEPT LINK**

OR-C4 | Primary home: RX-ST02

PIPELINE HELPER

P2 → P3 → electron count. Remember atom count for I_2 .

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help**H1**

H1: I changes $-1 \rightarrow 0$; S changes $-2 \rightarrow 0$.

H2

H2: Two I^- form I_2 , so iodine loses $2e^-$ total; S^{2-} loses $2e^-$.

STOP CHECK

If H2 makes the representation clear, finish independently before looking at Appendix A.

PYQ

Try first - choose your pipeline stopping point

Which species cannot function as an oxidising agent?

A SO_4^{2-} B MnO_4^- C N^{3-} D BrO_3^- [OPEN ORIGINAL EXAMSIDE SOURCE](#)**CONCEPT LINK**

AR-C2 / AR-C6 | Primary home: RX-ST03

PIPELINE HELPER

Use SELF logic: OA must be reducible.

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help**H1**

H1: An oxidising agent must itself be able to undergo reduction.

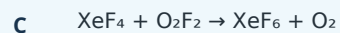
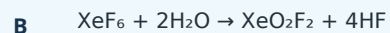
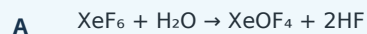
H2H2: N in N^{3-} is already at -3 in the source-context range; it cannot be reduced further.**STOP CHECK**

If H2 makes the representation clear, finish independently before looking at Appendix A.

PYQ

Try first - choose your pipeline stopping point

Which xenon-fluorine reaction is redox?



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RR-C2 / RR-C6 + ON-C4 | Primary home: RX-ST04

PIPELINE HELPER

Exception first: oxygen bonded to F is not -2. Then run P2-P4.

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help

H1

H1: Track Xe in all options; in option C also track oxygen in O_2F_2 .

H2

H2: In C, Xe $+4 \rightarrow +6$. The source gives O = +1 in O_2F_2 .

H3

H3: O in O_2F_2 changes $+1 \rightarrow 0$ in O_2 , giving a DOWN change paired with Xe UP.

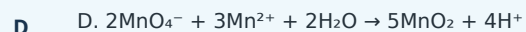
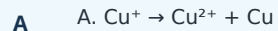
STOP CHECK

Can you finish now? If yes, do not open Appendix A.

PYQ

Try first - choose your pipeline stopping point

Which listed reactions are disproportionation reactions?



OPEN ORIGINAL EXAMSIDE SOURCE

CONCEPT LINK

RT-C4 / RT-C5 / RT-C6 | Primary home: RX-ST05

PIPELINE HELPER

Count starting ONs of the same element: one = split candidate; two = converge candidate.

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help

H1

H1: One starting ON must split both lower and higher.

H2

H2: A: Cu +1 $\rightarrow$ +2/0. B: Mn +6 $\rightarrow$ +7/+4. C: Mn only moves down; O moves up. D: Mn +7/+2 converge to +4.

H3

H3: A and B split. D converges, so it is comproportionation.

STOP CHECK

Can you finish now? If yes, do not open Appendix A.

PYQ

Try first - choose your pipeline stopping point

For $2[\text{Au}(\text{CN})_2]^- + \text{Zn} \rightarrow 2\text{Au} + [\text{Zn}(\text{CN})_4]^{2-}$, which labels are correct: redox, displacement, decomposition, combination?

FIRST MOVE

Decide whether this needs only oxidation-number work, electron accounting, agent logic, or full type classification.

[OPEN ORIGINAL EXAMSIDE SOURCE](#)

CONCEPT LINK

RT-C3 + RR-C2 | Primary home: RX-ST05

PIPELINE HELPER

Answer two independent questions: redox status and topology.

ExamSIDE Cumulative - Subtopic 06 | original source linked

Progressive help

H1

H1: $\text{Zn } 0 \rightarrow +2$; $\text{Au } +1 \rightarrow 0$.

H2

H2: That proves redox; Zn also replaces Au, so displacement. Not decomposition or combination.

STOP CHECK

If H2 makes the representation clear, finish independently before looking at Appendix A.

APP A

QUESTION → MINIMUM P1-P7 PREFIX → ANSWER → CHECK

WHY MINIMUM-PATH SOLUTIONS?

Integrated mastery means knowing how much reasoning is necessary. A good solution should not execute agent naming when the question asks only for oxidation number.

APP A

C01 RX-ST01

Answer: 19

- 1 $\text{K}_2\text{Cr}_2\text{O}_7$ gives $\text{Cr} = +6$.
- 2 KMnO_4 gives $\text{Mn} = +7$.
- 3 K_2FeO_4 gives $\text{Fe} = +6$.
- 4 Total = 19.

C02 RX-ST02

Answer: 4

- 1 Each I^- loses 1e^- ; two iodides give $X = 2$.
- 2 S^{2-} loses 2e^- , so $Y = 2$.
- 3 $X + Y = 4$.

APP A

C03 RX-ST03

Answer: N^{3-}

- 1 An OA must gain electrons / be reduced.
- 2 N in N^{3-} is already -3.
- 3 It cannot be reduced further in this source framework.
- 4 Therefore N^{3-} cannot function as the OA.

C04 RX-ST04

Answer: $\text{XeF}_4 + \text{O}_2\text{F}_2$ $\text{XeF}_6 + \text{O}_2$

- 1 A and B keep Xe at +6.
- 2 D keeps Xe at +2.
- 3 In C, Xe $+4 \rightarrow +6$ while O $+1 \rightarrow 0$.
- 4 Therefore C is redox.

APP A

C05 RX-ST05

Answer: A and B

- 1 A is disproportionation.
- 2 B is disproportionation.
- 3 C is intramolecular redox/decomposition, not disproportionation of Mn.
- 4 D is comproportionation.
- 5 Therefore A and B.

C06 RX-ST05

Answer: Redox and displacement only

- 1 Zn is oxidised; Au is reduced, so redox.
- 2 Zn replaces Au from the complex species, so displacement.
- 3 Not decomposition or combination.

Cumulative-transfer self-audit

This book must not alter canonical primary coverage.

AUDIT

Cumulative sample

SELECTED 6 / PRESENT 6

C01 RX-ST01; C02 RX-ST02; C03 RX-ST03; C04 RX-ST04; C05 RX-ST05; C06 RX-ST05.

NEW_PRIMARY_REQUIRED

0

CUMULATIVE_SELECTED

6

CUMULATIVE_PRESENT

6

PRIMARY_HOME_CHANGED

0

DUPLICATE_PRIMARY_PLACEMENTS

0

HINT_FAILURES

0

SOLUTION_FAILURES

0

SOURCE_LINK_FAILURES

0

Interpretation

WHY THIS IS NOT DOUBLE COUNTING

Canonical coverage asks: where does each eligible question live? Each C01-C06 still lives in its original RX-ST01-05 transfer book. Subtopic 06 only reuses them for mixed practice.

WHAT THIS DOES TEST

Whether the learner can identify the question family, choose the correct pipeline stopping point, and transfer across subtopics without being told which concept to use.

STATUS: PASS

CUMULATIVE_STATUS = PASS. Primary coverage impact = 0; cumulative support = complete.

REDOX EXAMSIDE FULL-CORPUS AUDIT

Frozen snapshot: 29 Numerical + 45 MCQ = 74 candidates

Scope classifier uses the supplied Redox reaction.pdf as the hard chemistry boundary

ELIGIBLE_IN_SCOPE

24 candidates
must have one canonical primary home

PARTIAL_SCOPE

8 candidates
contain an in-scope fragment but need
outside knowledge

OUT_OF_SCOPE

42 candidates
depend mainly on excluded chemistry

Important correction from the full audit: three additional eligible disproportionation PYQs were not in the first RX-ST05 transfer build.

They are backfilled in redox_subtopic_05_examside_questions_v2.pdf before chapter closeout.

RULES

Eligible

ELIGIBLE_IN_SCOPE

The question can be solved using only oxidation-number rules, simple electron accounting, oxidation/reduction, OA/RA logic, redox recognition, or reaction-type ideas explicitly inside the supplied PDF.

Partial

PARTIAL_SCOPE

An in-scope Redox step is present, but the answer also requires chemistry the source does not teach - e.g. reagent strength trends, acid-name formulas, accessible-state catalogues, or solution concentration arithmetic.

Excluded

OUT_OF_SCOPE

Minimum path depends mainly on titration/normality/equivalent weight, full redox balancing, medium-specific product chemistry, salt/qualitative analysis, coordination compounds, acid-base theory, or reaction-specific inorganic knowledge.

CANONICAL PRIMARY RULE

Every eligible item has exactly one primary subtopic. Duplicate-reasoning PYQs still need placement; they may add practice weight without creating a new concept.

Candidate ledger 1-10

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
N01	2026-04-06	KMnO ₄ /KI then thiosulfate titration	OUT_OF_SCOPE	redox titration + molarity stoichiometry
N02	2026-01-28	KI + KMnO ₄ ; iodine titrated with thiosulfate	OUT_OF_SCOPE	iodometric titration + solution stoichiometry
N03	2026-01-24	electron count: I ⁻ →I ₂ and S ²⁻ →S	ELIGIBLE_IN_SCOPE	RX-ST02
N04	2026-01-23	dichromate required for Mohr's salt	OUT_OF_SCOPE	quantitative redox stoichiometry / titration
N05	2025-04-04	Mn ON change + ferric acetate d-electron count	PARTIAL_SCOPE	Mn change is in scope; ferric acetate test/d-electrons are external
N06	2025-01-22	CO ₂ + Ca(OH) ₂ then HCl neutralisation	OUT_OF_SCOPE	gas/acid-base quantitative stoichiometry
N07	2024-04-04	KMnO ₄ vs oxalic acid titration molarity	OUT_OF_SCOPE	redox titration
N08	2024-01-31	Cr oxidation state in K ₂ Cr ₂ O ₇ /KCl/H ₂ SO ₄ product	OUT_OF_SCOPE	requires chromyl-chloride/product knowledge outside source
N09	2024-01-31	moles H ⁺ needed for MnO ₄ ⁻ /oxalate	OUT_OF_SCOPE	full balancing / stoichiometric coefficients

Candidate ledger 11-20

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
N11	2024-01-30	balance $\text{MnO}_4^- + \text{I}^-$ in alkaline medium	OUT_OF_SCOPE	full medium-dependent redox balancing
N12	2024-01-29	Fe oxidation number in brown-ring compound	OUT_OF_SCOPE	coordination/brown-ring-test identity outside source
N13	2023-04-13	$\text{Cr}_2\text{O}_7^{2-} + \text{H}^+ + \text{Fe}^{2+}$ coefficient sum	OUT_OF_SCOPE	full redox balancing
N14	2023-04-10	electron gain by permanganate in alkaline medium	OUT_OF_SCOPE	requires medium-specific Mn product knowledge not taught
N15	2023-04-08	sum ON in $\text{Fe}(\text{CO})_5$, VO^{2+} , WO_3	ELIGIBLE_IN_SCOPE	RX-ST01
N16	2023-04-06	Mo oxidation state in ammonium phosphomolybdate	OUT_OF_SCOPE	compound/salt-analysis formula knowledge outside source
N17	2023-02-01	Br ON in bromic/perbromic acid	PARTIAL_SCOPE	ON arithmetic is in scope; acid-name→formula knowledge is not supplied
N18	2023-01-31	P oxidation state in hypophosphoric acid	PARTIAL_SCOPE	ON arithmetic is in scope; acid-name→formula knowledge is not supplied
N19	2023-01-25	strong acid density/neutralisation volume	OUT_OF_SCOPE	acid-base quantitative stoichiometry

Candidate ledger 21-30

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
N21	2022-07-26	CuSO ₄ iodometric titration	OUT_OF_SCOPE	iodometric titration
N22	2022-06-28	KMnO ₄ titration of Mohr's salt	OUT_OF_SCOPE	redox titration
N23	2022-06-25	acid basicity from neutralisation	OUT_OF_SCOPE	acid-base stoichiometry
N24	2021-07-27	oxalic acid vs KMnO ₄ solution strength	OUT_OF_SCOPE	redox titration
N25	2021-02-26	dichromate treated with base; Cr in product	OUT_OF_SCOPE	requires chromate/dichromate equilibrium product identity
N26	2021-02-25	NaOH/Na ₂ CO ₃ double-indicator titration	OUT_OF_SCOPE	acid-base titration
N27	2020-09-05	dichromate volume for ferrous oxalate	OUT_OF_SCOPE	quantitative redox stoichiometry
N28	2020-09-04	Na ₂ CO ₃ ·xH ₂ O from normality	OUT_OF_SCOPE	normality / hydrate stoichiometry
N29	2020-09-02	ON in K ₂ Cr ₂ O ₇ , KMnO ₄ , K ₂ FeO ₄	ELIGIBLE_IN_SCOPE	RX-ST01

Candidate ledger 31-40

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
M02	2026-04-02	halogen oxidising power + layer test	PARTIAL_SCOPE	displacement idea is in scope; halogen power trend/layer test is external
M03	2026-01-24	Cl ₂ into cold KOH; final concentrations	PARTIAL_SCOPE	source gives disproportionation equation; concentration stoichiometry is outside
M04	2026-01-23	Cr state after KI + acidified dichromate	OUT_OF_SCOPE	requires reagent-specific reduction product knowledge
M05	2025-01-29	KI/KIO ₃ stoichiometry + equivalent weight	OUT_OF_SCOPE	equivalent weight / solution stoichiometry
M06	2025-01-28	KMnO ₄ titration heating/catalysis statements	OUT_OF_SCOPE	redox titration procedure and catalysis
M07	2025-01-28	match combination/decomposition/displacement/disproportionation	ELIGIBLE_IN_SCOPE	RX-ST05
M08	2025-01-28	oxidations done by both dichromate and permanganate	OUT_OF_SCOPE	reagent-specific oxidation capability beyond supplied equations
M09	2025-01-22	chlorine oxyanion that does not disproportionate	ELIGIBLE_IN_SCOPE	RX-ST05
M10	2024-04-08	thiosulfate with I ₂ vs Br ₂ ; justify dual behaviour	PARTIAL_SCOPE	ON tracking is in scope; stronger-oxidant comparison Br ₂ vs I ₂ is external

Candidate ledger 41-50

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
M12	2024-04-06	match four redox reaction types	ELIGIBLE_IN_SCOPE	RX-ST05
M13	2024-04-04	MnO ₂ /H ₂ SO ₄ salt gives greenish-yellow gas	OUT_OF_SCOPE	salt analysis / reaction-specific product knowledge
M14	2024-02-01	which reactions are disproportionation	ELIGIBLE_IN_SCOPE	RX-ST05
M15	2024-02-01	acidic dichromate half-reaction X,Y,Z,A	OUT_OF_SCOPE	full half-reaction balancing
M16	2024-01-29	balance chlorine disproportionation coefficients	OUT_OF_SCOPE	balancing algorithm is outside current source teaching contract
M17	2024-01-29	alkaline permanganate oxidises I ⁻ to ?	OUT_OF_SCOPE	medium-specific reagent/product chemistry
M18	2024-01-27	species that cannot function as oxidising agent	ELIGIBLE_IN_SCOPE	RX-ST03
M19	2023-04-08	redox vs acid-base titration indicator statements	OUT_OF_SCOPE	titration indicators / oxidation potential
M20	2023-04-08	IO ₃ ⁻ + I ⁻ + H ⁺ coefficient x	OUT_OF_SCOPE	redox balancing

Candidate ledger 51-60

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
M22	2022-07-29	weak-acid indicator X	OUT_OF_SCOPE	acid-base indicators
M23	2022-07-26	phenolphthalein assertion/reason	OUT_OF_SCOPE	acid-base indicator theory
M24	2022-07-26	which reaction is NOT disproportionation	ELIGIBLE_IN_SCOPE	RX-ST05
M25	2022-06-26	which reaction IS disproportionation	ELIGIBLE_IN_SCOPE	RX-ST05
M26	2021-07-27	P states in $\text{H}_4\text{P}_2\text{O}_7/\text{H}_4\text{P}_2\text{O}_5/\text{H}_4\text{P}_2\text{O}_6$	ELIGIBLE_IN_SCOPE	RX-ST01
M27	2021-07-25	process with oxidation-state change of five	ELIGIBLE_IN_SCOPE	RX-ST01
M28	2021-03-18	order N states in $\text{NO}/\text{NO}_2/\text{N}_2\text{O}/\text{NO}_3^-$	ELIGIBLE_IN_SCOPE	RX-ST01
M29	2021-03-16	NaH oxidising-agent statement + pyridine basicity	PARTIAL_SCOPE	NaH redox logic is in scope; pyridine basicity is outside
M30	2020-09-02	HCl/NaOH titration equipment	OUT_OF_SCOPE	acid-base titration

Candidate ledger 61-70

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
M32	2020-01-07	identify redox reaction among options	ELIGIBLE_IN_SCOPE	RX-ST04
M33	2020-01-07	K oxidation number in $K_2O/K_2O_2/KO_2$	ELIGIBLE_IN_SCOPE	RX-ST01
M34	2019-04-12	example of disproportionation reaction	ELIGIBLE_IN_SCOPE	RX-ST05
M35	2019-04-09	order N states in $NO/N_2O/NO_2/N_2O_3$	ELIGIBLE_IN_SCOPE	RX-ST01
M36	2019-04-08	moles $KMnO_4$ for Fe/oxalate mixture	OUT_OF_SCOPE	quantitative redox stoichiometry
M37	2019-01-12	oxalic acid neutralises NaOH	OUT_OF_SCOPE	acid-base stoichiometry
M38	2019-01-12	water hardness in equivalents	OUT_OF_SCOPE	equivalents/water chemistry
M39	2019-01-10	electrons per CO_2 from oxalate oxidation	ELIGIBLE_IN_SCOPE	RX-ST02
M40	2017-01-17	which reaction is redox (Xe/O-F option)	ELIGIBLE_IN_SCOPE	RX-ST04

Candidate ledger 71-74

74-item frozen ExamSIDE snapshot

LEDGER

ID	Date	Candidate	Scope status	Primary / exclusion reason
M42	2014	H ₂ O ₂ acts as reducing agent in which half-reactions	ELIGIBLE_IN_SCOPE	RX-ST03
M43	2013	balance permanganate/oxalate coefficients	OUT_OF_SCOPE	full redox balancing
M44	AIEEE 2006	reaction showing oxidising behaviour of H ₂ SO ₄	ELIGIBLE_IN_SCOPE	RX-ST03
M45	AIEEE 2002	which reaction is redox	ELIGIBLE_IN_SCOPE	RX-ST04

Backfill discovered by the full-corpus pass

Question-family duplicates still count as required coverage.

BACKFILL

Three missing eligible questions

M24 -> RX-ST05 -> E06

JEE Main 2022 - 26 Jul Morning

Which reaction is NOT disproportionation?

Status: ELIGIBLE_IN_SCOPE; architecture novelty: known/duplicate disproportionation family; coverage action: BACKFILL REQUIRED.

M25 -> RX-ST05 -> E07

JEE Main 2022 - 26 Jun Evening

Which reaction IS disproportionation?

Status: ELIGIBLE_IN_SCOPE; architecture novelty: known/duplicate disproportionation family; coverage action: BACKFILL REQUIRED.

M34 -> RX-ST05 -> E08

JEE Main 2019 - 12 Apr Morning

Example of disproportionation reaction

Status: ELIGIBLE_IN_SCOPE; architecture novelty: known/duplicate disproportionation family; coverage action: BACKFILL REQUIRED.

WHY THIS MATTERS

A duplicate reasoning pattern may not justify a new teaching page, but it still belongs in the transfer corpus. Full-corpus enumeration therefore happens before final chapter acceptance.

Eligible corpus -> canonical primary homes

After the RX-ST05 backfill, every eligible item is uniquely placed.

PLACEMENT

24 eligible questions

RX-ST01 Oxidation Number

7/7

RX-ST02 Oxidation & Reduction

2/2

RX-ST03 Oxidising / Reducing Agents

4/4

RX-ST04 Redox vs Non-redox

3/3

RX-ST05 Types of Redox Reactions

8/8

CHAPTER EXTERNAL-CORPUS STATUS

ELIGIBLE_TOTAL = 24; PLACED_UNIQUE = 24.
MISSING = 0; DUPLICATE_PRIMARY_PLACEMENTS = 0.

Non-eligible corpus

PARTIAL_SCOPE = 8

Retained with explicit outside dependencies. They do not shape learner architecture and are not silently rewritten as original PYQs.

OUT_OF_SCOPE = 42

Predominantly titration/normality, balancing, acid-base, salt analysis, coordination chemistry, medium-dependent products or reagent-specific inorganic knowledge.

RX-ST06

Integrated Decision Pipeline has no new primary PYQ denominator. Its cumulative questions retain their RX-ST01-05 homes.